Circadian Period Q10 Calculator

User manual for the web app

App: dxda6216.github.io/q10_two_temperatures/index.html
Source: github.com/dxda6216/q10_two_temperatures · License: GPL-3.0

Contents

1. Introduction
2. Opening the app
3. The screen at a glance
4. Entering your data
5. Reading the results
6. The plot
7. Exporting data to Excel
8. Messages and troubleshooting
9. How the calculations work
10. Tips and limitations

1 Introduction

The Circadian Period Q10 Calculator computes the temperature coefficient (Q10) of a circadian rhythm's period from measurements taken at two temperatures. It is a browser version of the Colab notebook `circadian_period_q10_2_temperatures.ipynb`, extended so you can enter several replicate period measurements at each temperature.

Q10 describes how much a rate changes for a 10 °C rise in temperature. For a rhythm, the rate is the frequency (1/period), so a Q10 near 1 means the period barely changes with temperature, the hallmark of *temperature compensation* in circadian clocks.

The app gives you:

- Q10 calculated from the mean period at each temperature, with an approximate standard error and 95% range when you have replicates.
- Summary statistics (n, mean, SD, SEM) for each temperature.
- A plot of the replicates, means ± SD, and the fitted exponential curve.
- Tab-delimited tables ready to paste into Excel.

> **Your data stays private**
> Every calculation runs inside your browser. Nothing you type is uploaded or saved, and closing or reloading the page discards it.

2 Opening the app

Online (recommended)

Open https://dxda6216.github.io/q10_two_temperatures/index.html in any modern browser (Chrome, Edge, Firefox or Safari). Nothing needs to be installed.

Offline

1. Open `index.html` in the GitHub repository and click **Download raw file**.
2. Double-click the downloaded file to open it in your browser. It is fully self-contained and works without an internet connection (using system fonts).

3 The screen at a glance

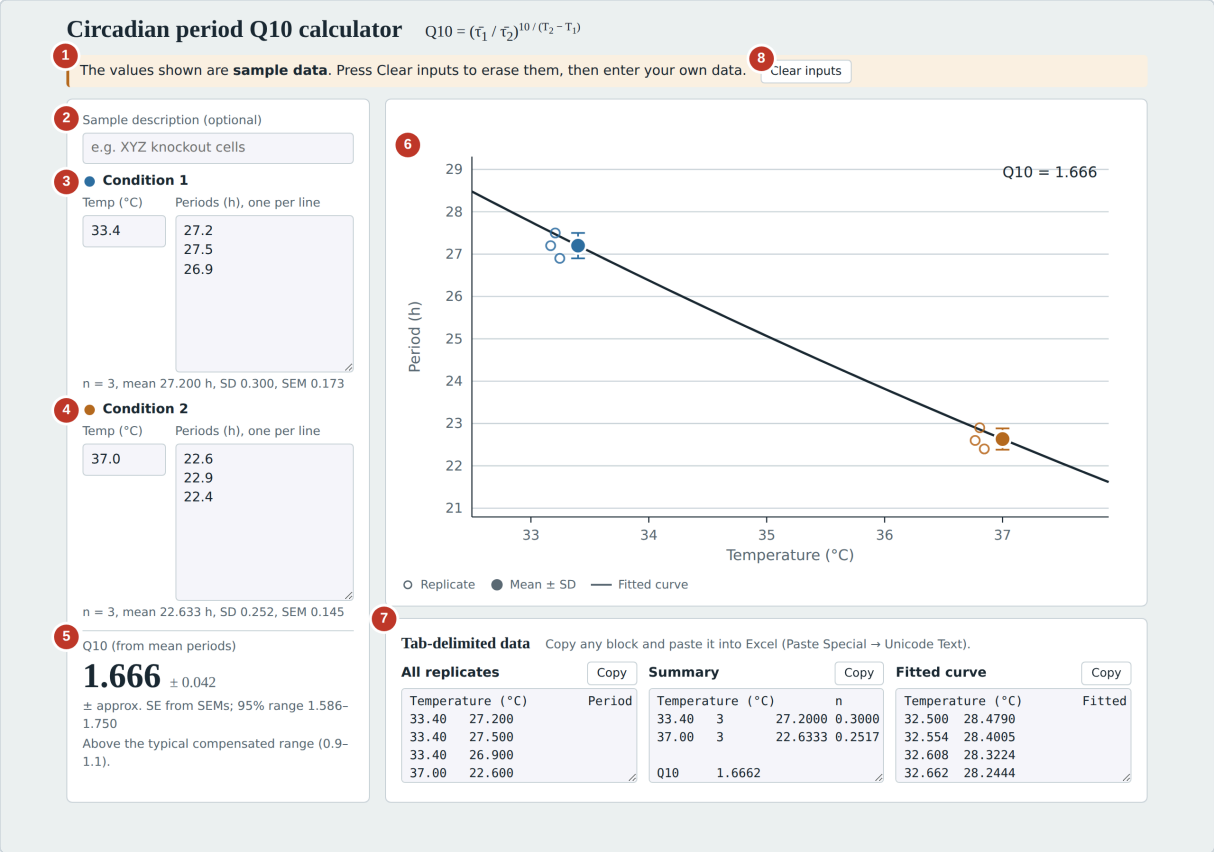


Figure 1. The app as first opened, showing sample data. On screens narrower than about 760 px the panels stack vertically in the same order.

#	Area	What it does
1	Sample-data notice	Reminds you that the values shown on first load are an example. After you clear them, it changes to a short instruction.
8	**Clear inputs** button	Empties every input field at once so you can start fresh.
2	Sample description	Optional label, displayed as the plot title.
3	Condition 1 (blue)	Temperature and period measurements for the first temperature, with live summary statistics underneath.
4	Condition 2 (orange)	The same for the second temperature.
5	Result	Q10, its uncertainty, an interpretation, and any error messages.
6	Plot	Period versus temperature with replicates, means ± SD and the fitted curve.
7	Tab-delimited data	Three copyable tables for Excel: all replicates, summary, fitted curve.

4 Entering your data

1. **Clear the sample data.** Click **Clear inputs** in the notice bar at the top. All fields empty, and the cursor moves to the first temperature box.



The notice after clearing.

2. **(Optional) Describe the sample**, for example *Wild-type fibroblasts*. This becomes the plot title.
3. **Enter Condition 1.** Type the temperature in °C in *Temp*, then the measured periods in hours in *Periods*.
4. **Enter Condition 2** the same way. Either temperature may be the higher one.
5. **Read the result.** Everything updates as you type; there is no Calculate button.

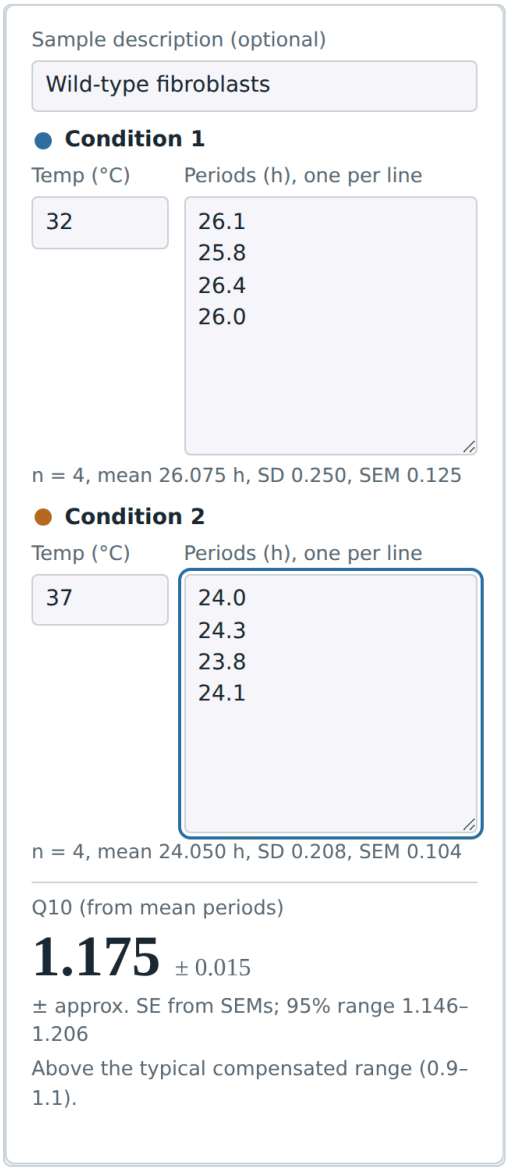


Figure 2. Inputs filled in with four replicates per temperature.

Entering replicate periods

Put one or more values in each *Periods* box. Separate them with any of: line breaks, spaces, commas, semicolons or tabs. You can paste a column copied straight from Excel. A single value per temperature is fine too; it reproduces the original two-point calculation.

Below each box the app shows *n*, the mean, and (when n ≥ 2) the SD and SEM, so you can check that every value was read correctly.

> **Check before you trust the result**
> If the n shown under a box doesn't match the number of measurements you meant to enter, look for a stray character or a missing separator.

5 Reading the results

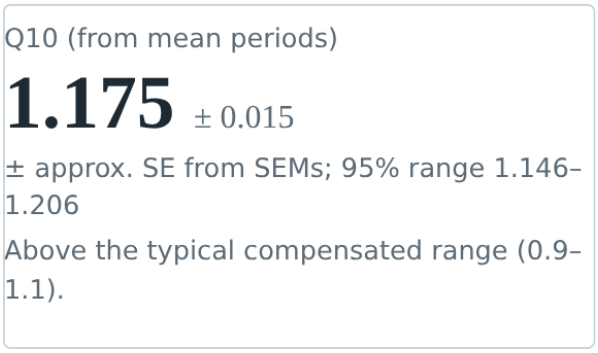


Figure 3. Result panel for the data in Figure 2.

The result panel shows three things:

- **Q10** in large type, calculated from the mean periods.
- **± value**: the approximate standard error of Q10, followed by an approximate 95% range. These appear only when both temperatures have at least two replicates.
- **Interpretation** of where Q10 falls relative to the typical temperature-compensated range.

Interpretation labels

Q10	Label shown	Meaning
0.9 – 1.1	Within the typical compensated range	Period is largely independent of temperature.
above 1.1	Above the typical compensated range	Period shortens noticeably as temperature rises.
below 0.9	Overcompensated	Period lengthens as temperature rises.

In the example (Figure 3), Q10 = 1.175 ± 0.015 with a 95% range of 1.146–1.206. Because the whole range lies above 1.1, the result is labelled *Above the typical compensated range*.

> **When there's no ± value**
> With only one value at either temperature, there is no spread from which to estimate uncertainty. If one condition has replicates and the other does not, the panel reminds you to add replicates at both temperatures.

6 The plot

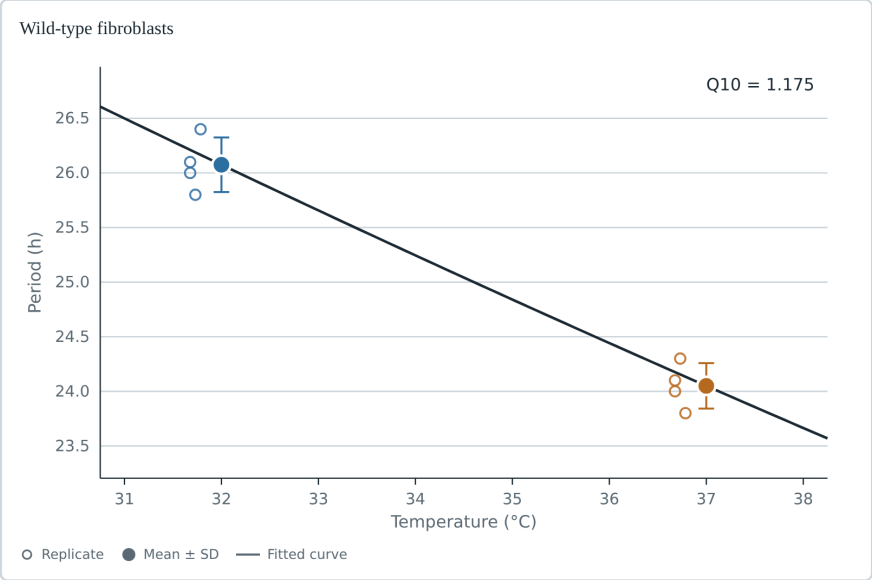


Figure 4. Plot for the data in Figure 2.

Element	Meaning
Open circles	Individual replicate periods, drawn slightly left of the mean so they don’t hide it.
Filled circle with bars	Mean period at that temperature, with ± SD error bars when n ≥ 2. Blue is Condition 1, orange is Condition 2.
Line	Exponential curve implied by the Q10, passing through both means and extending 25% of the temperature span beyond each end.
Q10 label	The calculated Q10, shown in the top-right corner.

Hover the pointer over any circle to see its exact temperature and period.

7 Exporting data to Excel

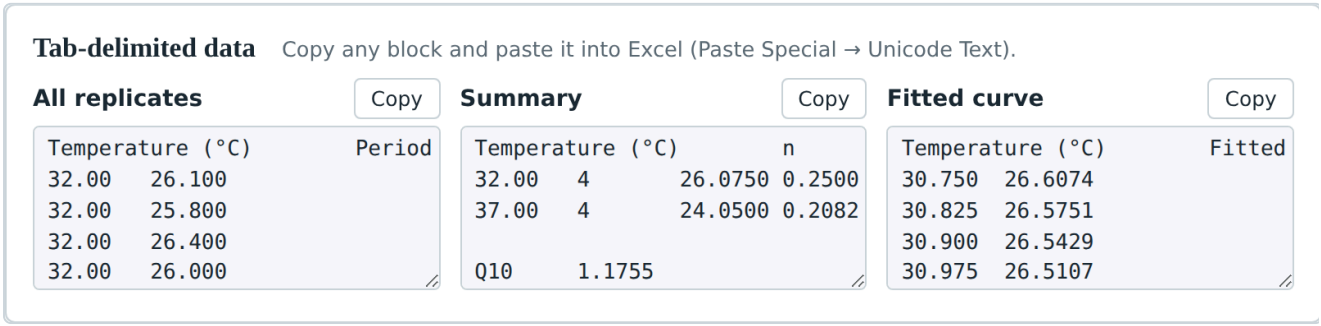


Figure 5. The Tab-delimited data panel.

Block	Columns
All replicates	Temperature (°C), Period (h): one row per measurement.
Summary	Temperature (°C), n, Mean period (h), SD, SEM for each condition, followed by Q10.
Fitted curve	Temperature (°C), Fitted period (h): 101 points along the curve.

1. Click **Copy** above the block you want. The button briefly changes to *Copied*.
2. In Excel, right-click the cell where the table should start and choose **Paste Special…**
3. Select **Unicode Text** and click **OK**. Each value lands in its own cell.

> **Recreating the plot in Excel**
> Paste *All replicates* and *Fitted curve* side by side, then make a scatter chart with the replicates as markers and the fitted curve as a smooth line.

8 Messages and troubleshooting

When something prevents a calculation, a red message appears in the result panel and the plot and tables are blanked until the problem is fixed.

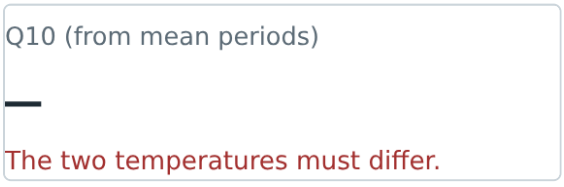


Figure 6. Message shown when both temperatures are the same.

Message	Cause	What to do
Fill in both temperatures.	A *Temp* box is empty (also shown right after clearing).	Enter a temperature for each condition.
Temperatures must be numbers.	A temperature isn't a valid number.	Use digits and a decimal point, e.g. `33.4`.
Not a number: …	A period box contains text the app can't read; the offending entries are listed.	Delete or correct those entries. Use a point, not a comma, as the decimal mark.
Enter at least one period for each temperature.	A *Periods* box is empty.	Add at least one value.
The two temperatures must differ.	Both temperatures are equal, so Q10 is undefined.	Correct one of the temperatures.
Periods must be greater than zero.	A period is zero or negative.	Check for typos such as a stray minus sign.

> **Decimal commas**
> Commas are treated as separators between values, so `27,2` is read as two values, 27 and 2. Always use a decimal point.

Other questions

The Copy button doesn't work. Some browsers block clipboard access, particularly when the file is opened locally. Click inside the block, press Ctrl + A (⌘ + A on Mac) then Ctrl + C, and paste as usual.

I want the sample data back. Reload the page.

Can I save my inputs? No; the app deliberately stores nothing. Keep your raw data in a spreadsheet and paste it in when needed.

9 How the calculations work

Q10

Replicates at each temperature are averaged. With mean periods $\bar{\tau}_1$ at T_1 and $\bar{\tau}_2$ at T_2:

```
Q10 = (τ1 / τ2) ^ (10 / (T2 − T1))
```

This follows from the general definition Q10 = $(R_2/R_1)^{10/(T_2-T_1)}$ with rate R = 1/τ.

Fitted curve

```
τ(T) = τ1 · Q10 ^ (−(T − T1) / 10)
```

The curve passes through both mean periods and is evaluated at 101 evenly spaced temperatures, from 25% of the temperature span below the lower temperature to 25% above the higher one.

Summary statistics

For each temperature: n; the mean; the sample standard deviation SD (n − 1 denominator); and the standard error of the mean, SEM = SD / √n.

Uncertainty of Q10

When both temperatures have at least two replicates, each mean's SEM is propagated to ln Q10 (delta method):

```
SE(ln Q10) = (10 / |T2 − T1|) · √[ (SEM1 / τ1)² + (SEM2 / τ2)² ]
SE(Q10)   ≈ Q10 · SE(ln Q10)
95% range ≈ Q10 · exp(±1.96 · SE(ln Q10))
```

Working on the log scale keeps the range positive and makes it slightly asymmetric around Q10, which reflects Q10's ratio form.

10 Tips and limitations

- **The uncertainty is approximate.** It uses a normal (1.96) multiplier, which understates uncertainty when n is small. For formal reporting with few replicates, consider a t-based interval or a bootstrap analysis.
- **Temperatures far apart give a more precise Q10.** Because ΔT appears in the exponent, a small temperature difference magnifies measurement error. Note how 10/|ΔT| scales the standard error.
- **Only two temperatures.** The fitted curve assumes a single exponential relationship and cannot show non-linearity. For three or more temperatures, use the companion tool dxda6216/q10.
- **Units.** Temperatures are in °C and periods in hours. Q10 itself is unitless, so any consistent period unit gives the same Q10, but the plot and tables label periods as hours.
- **The 0.9–1.1 range is a convention**, not a strict cut-off. Interpret borderline values in light of the uncertainty.